

BRAC UNIVERSITY
CSE460- VLSI DESIGN
Spring '26
Section: 09 Assignment: 01

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Question-1: Sketch a transistor-level schematic for a compound CMOS logic gate for each of the following functions:

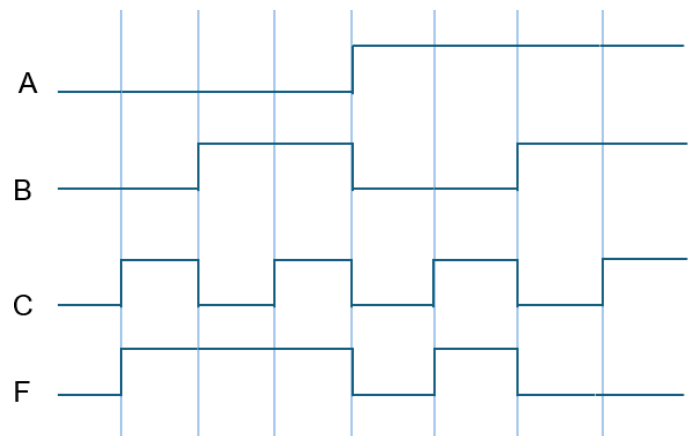
- a) $Y = (A \cdot B \cdot C + D)'$
- b) $Y = ((A \cdot B + C) \cdot D)'$
- c) $Y = (A \cdot B + C \cdot (A + B))'$

Question-2: A classroom has three temperature sensors to control the temperature of the room by controlling the air conditioner installed in that room. The temperature sensors produce a high output if the temperature crosses a certain threshold. The air conditioner turns on if the outputs of all three sensors are high, otherwise it remains off.

- a) Identify the logic circuit required to implement the system mentioned above and implement the circuit using 2:1 mux.
- b) Draw the circuit using transmission gates.

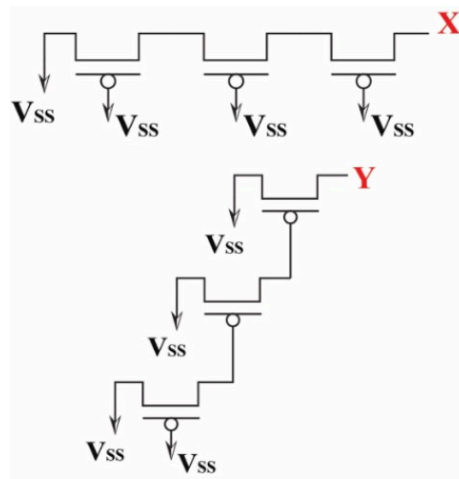
Question-3: For the given timing diagram,

- a) Synthesize the function $F(A,B,C)$.
- b) Draw the circuit using CMOS logic.
- c) Implement the circuit using a 4:1 MUX.

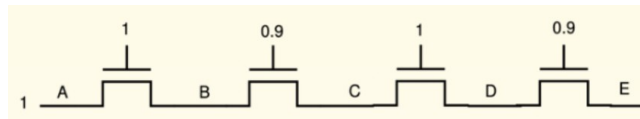


Question-4: Explain restoring and non-restoring properties of a device with the help of different kinds of CMOS tristate devices.

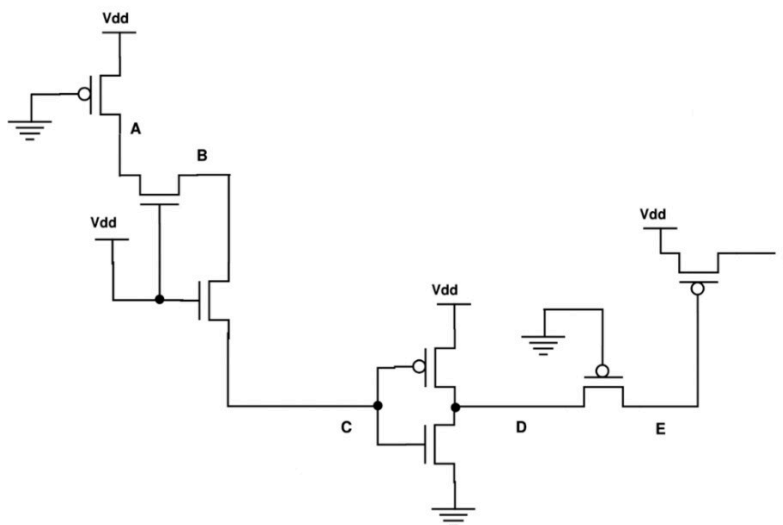
Question-5: From the following figure, find the value of the voltages at points X and Y if $V_{SS} = 0\text{ V}$ and $V_{tp} = -0.3\text{ V}$. Why do X and Y differ?



Question-6: From the following figure, find the voltages at each of the nodes, A, B, C, D, E. Assume, $V_{dd} = 1\text{ V}$, $Gnd = 0\text{ V}$, $V_{tn} = 0.2\text{ V}$ and $|V_{tp}| = 0.2\text{ V}$.



Question-7: From the following figure, find the voltages at each of the nodes, A, B, C, D, E and F. Assume, $V_{dd} = 5\text{ V}$, $Gnd = 0\text{ V}$, $V_{tn} = 0.5\text{ V}$ and $|V_{tp}| = 1.5\text{ V}$.



Question-8: From the following figure, find the voltages V_{out1} , V_{out2} and V_{out3} . Assume, $V_{tn} = 0.5\text{ V}$ and $|V_{tp}| = 1.5\text{ V}$.

